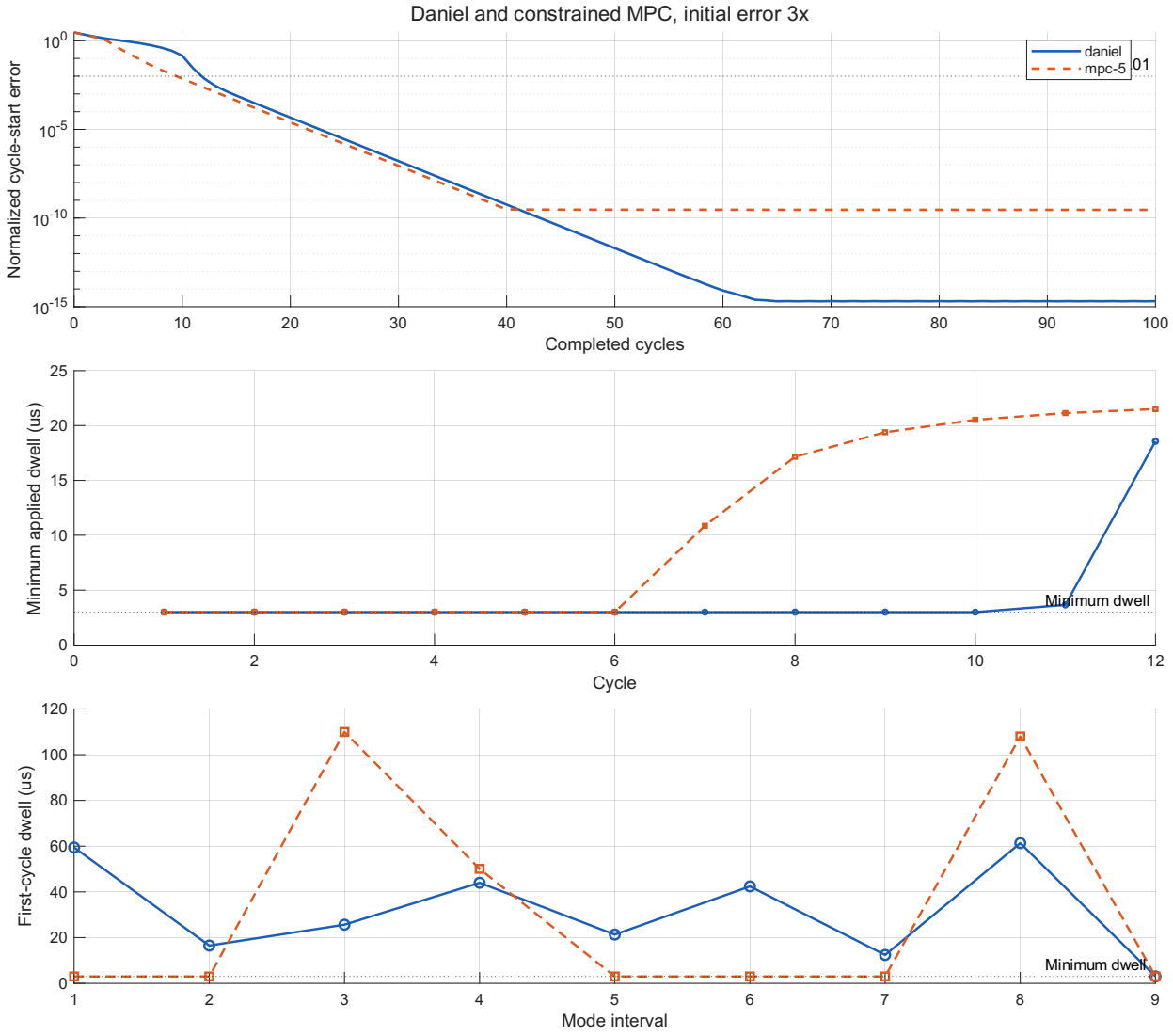
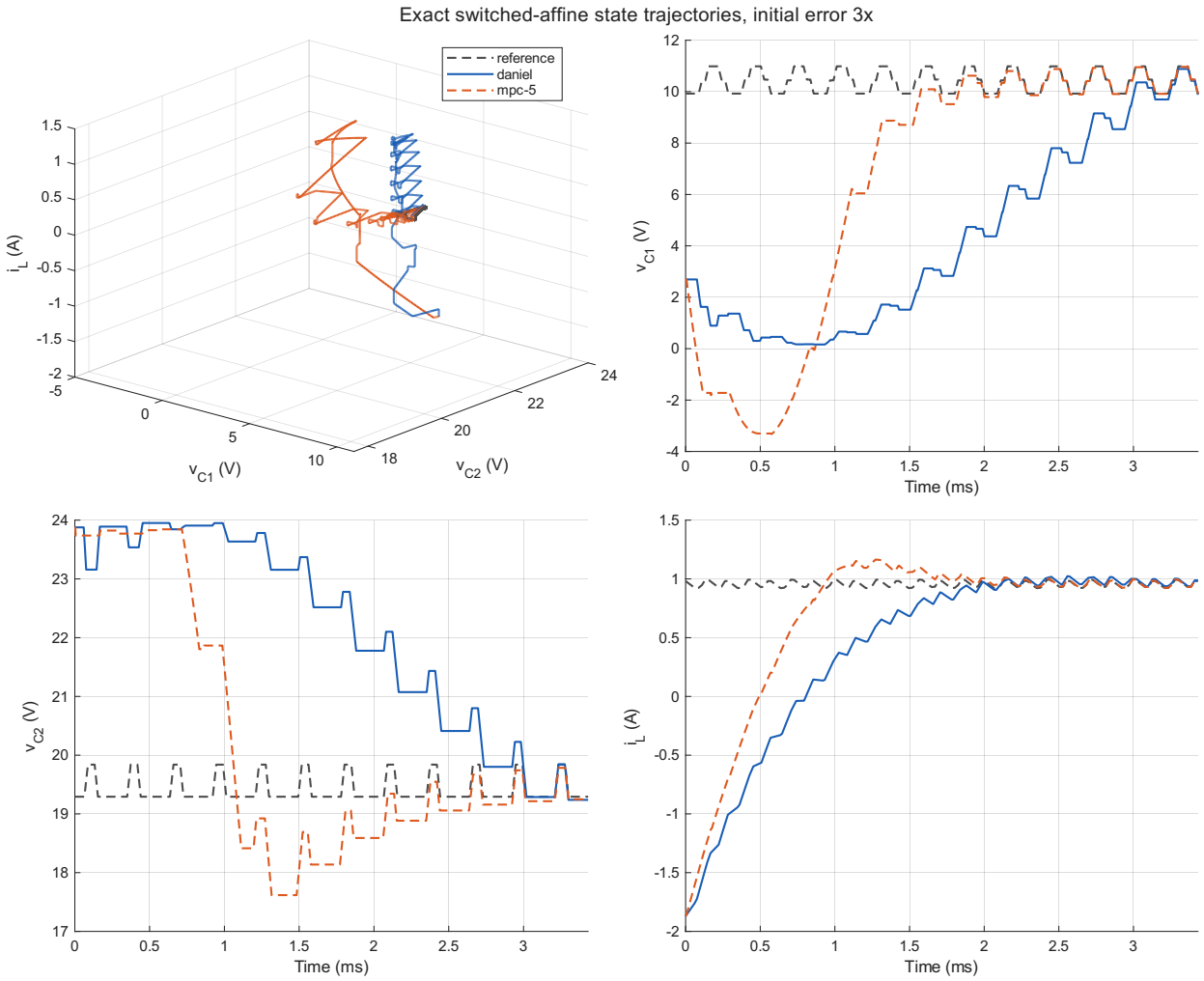
Daniel versus MPC

The threefold initial-error trial uses the same converter, LQR weights, 286 μs period, and 3 μs dwell bound. Blue denotes `daniel` and dashed orange denotes `mpc-5`; the MPC suffix is its prediction horizon in cycles. The panels show cycle-start error, minimum applied dwell, and first-cycle dwells.



Exact state trajectories over the first twelve nominal periods compare Daniel and five-cycle MPC with the dashed reference. MPC-5 settles earlier than Daniel but produces negative capacitor voltage. This trial also starts with negative current; it tests the mathematical model, not validated hardware operation. No nonlinear stability guarantee follows from these plots.



Computation time

Recorded MATLAB times per successful cycle include Daniel's LQR command and dwell conditioning, or MPC's QP solve. The table covers all three initial-error scales; each completed run supplies 100 calls. First-call overhead is included. Offline design, state normalization, plant propagation, and plotting are excluded. These are desktop timings, not ESP32 deadline measurements.

Scale	Method	Calls	Median (μs)	95th percentile (μs)
1x	daniel	100	12.70	145.15
1x	mpc-5	100	265.75	2200.20
2x	daniel	100	12.20	17.05
2x	mpc-5	100	262.20	688.90
3x	daniel	100	9.35	18.05
3x	mpc-5	100	301.40	972.15

The 95th percentile describes the slower tail of the recorded calls, not a worst-case execution bound. Missing entries mean no successful MPC calls; the failed solve is reported separately below the table.